

Data Structures Home Work Set 3

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1. Write a C program to store n elements in an array using dynamic memory allocation and find largest number.

Program:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {  
    int n, i, max;  
  
    int *arr;  
  
    printf("Enter number of elements: ");  
  
    scanf("%d", &n);  
  
    arr = (int*)malloc(n * sizeof(int));  
  
    printf("Enter %d elements:\n", n);  
  
    for(i = 0; i < n; i++) {  
        scanf("%d", &arr[i]);  
    }  
}
```

```

    max = arr[0];

    for(i = 1; i < n; i++) {

        if(arr[i] > max) {

            max = arr[i];

        }

    }

    printf("The largest number is: %d\n", max);

    free(arr);

    return 0;

}

```

Output:

Enter number of elements: 3

Enter 3 elements:

12

65

39

The largest number is: 65

2. Print an array in reverse order using pointers.

Program:

```
#include <stdio.h>
```

```
int main() {
```

```

int n, i;

int arr[100];

int *ptr;

printf("Enter number of elements: ");

scanf("%d", &n);

ptr = arr;

printf("Enter the elements:\n");

for(i = 0; i < n; i++) {

    scanf("%d", ptr + i);

}

ptr = &arr[n - 1];

printf("The reversed array is:\n");

for(i = 0; i < n; i++) {

    printf("%d ", *ptr);

    ptr--;

}

printf("\n");

return 0;

}

```

Output:

Enter number of elements: 3

Enter the elements:

43

76

The reversed array is:

76 43 12

3. Write a C program to store information of students using structure. Sort them by Name.

Program:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
struct Student {
```

```
    char name[50];
```

```
    int roll;
```

```
};
```

```
int main() {
```

```
    int n, i, j;
```

```
    struct Student s[100], temp;
```

```
    printf("Enter number of students: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++) {
```

```
        printf("Enter name and roll for student %d: ", i + 1);
```

```
        scanf("%s %d", s[i].name, &s[i].roll);
```

```
    }
```

```
    for(i = 0; i < n - 1; i++) {
```

```

    for(j = i + 1; j < n; j++) {
        if(strcmp(s[i].name, s[j].name) > 0) {
            temp = s[i];
            s[i] = s[j];
            s[j] = temp;
        }
    }
}

printf("Sorted students by name:\n");

for(i = 0; i < n; i++) {
    printf("Name: %s, Roll: %d\n", s[i].name, s[i].roll);
}

return 0;
}

```

Output:

Enter number of students: 3

Enter name and roll for student 1: SENTHIL

234

Enter name and roll for student 2: Senthil

245

Enter name and roll for student 3: sak

248

Sorted students by name:

Name: SENTHIL, Roll: 234

Name: Senthil, Roll: 245

Name: sak, Roll: 248

4. Write a C program to add 'n' distances given in inch-feet system using structures.

Program:

```
#include <stdio.h>
```

```
struct Distance {
```

```
    int feet;
```

```
    float inch;
```

```
};
```

```
int main() {
```

```
    int n, i;
```

```
    struct Distance d, sum = {0, 0.0};
```

```
    printf("Enter number of distances: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++) {
```

```
        printf("Enter distance %d in feet and inches: ", i + 1);
```

```
        scanf("%d %f", &d.feet, &d.inch);
```

```
        sum.feet = sum.feet + d.feet;
```

```
        sum.inch = sum.inch + d.inch;
```

```
    }
```

```

while(sum.inch >= 12.0) {

    sum.inch = sum.inch - 12.0;

    sum.feet = sum.feet + 1;

}

printf("The total added distance is: %d feet %.1f inches\n", sum.feet, sum.inch);

return 0;

}

```

Output:

Enter number of distances: 2

Enter distance 1 in feet and inches: 24

54

Enter distance 2 in feet and inches: 35

3.4

The total added distance is: 63 feet 9.4 inches

5. Write a C program to add two complex numbers by passing structure to a function.

Program:

```
#include <stdio.h>
```

```
struct Complex {
```

```
    float real;
```

```
    float imag;
```

```
};
```

```
struct Complex add(struct Complex n1, struct Complex n2) {  
    struct Complex temp;  
    temp.real = n1.real + n2.real;  
    temp.imag = n1.imag + n2.imag;  
    return temp;  
}
```

```
int main() {  
    struct Complex n1, n2, result;  
    printf("Enter real and imaginary parts of first complex number: ");  
    scanf("%f %f", &n1.real, &n1.imag);  
    printf("Enter real and imaginary parts of second complex number: ");  
    scanf("%f %f", &n2.real, &n2.imag);  
    result = add(n1, n2);  
    printf("The added complex number is: %.1f + %.1fi\n", result.real, result.imag);  
    return 0;  
}
```

Output:

Enter real and imaginary parts of first complex number: 2.4

Enter real and imaginary parts of second complex number: 32

4.9

The added complex number is: $34.4 + 49.9i$

6. Write a C program to maintain inventory management system using structure.

Program:

```
#include <stdio.h>
```

```
struct Item {
```

```
    int id;
```

```
    char name[50];
```

```
    int quantity;
```

```
};
```

```
int main() {
```

```
    int n, i;
```

```
    struct Item inv[100];
```

```
    printf("Enter number of items: ");
```

```
    scanf("%d", &n);
```

```
    for(i = 0; i < n; i++) {
```

```
        printf("Enter ID, Name, and Quantity for item %d: ", i + 1);
```

```
        scanf("%d %s %d", &inv[i].id, inv[i].name, &inv[i].quantity);
```

```
    }
```

```
    printf("Inventory Details:\n");
```

```
for(i = 0; i < n; i++) {  
  
    printf("Item ID: %d, Name: %s, Quantity: %d\n", inv[i].id, inv[i].name,  
inv[i].quantity);  
  
}  
  
return 0;  
  
}
```

Output:

Enter number of items: 2

Enter ID, Name, and Quantity for item 1: 2026

pen

12

Enter ID, Name, and Quantity for item 2: 2027

pencil

23

Inventory Details:

Item ID: 2026, Name: pen, Quantity: 12

Item ID: 2027, Name: pencil, Quantity: 23

7. Write a C program for decimal to binary, octal and hexa decimal conversion.

Program:

```
#include <stdio.h>
```

```

int main() {

    int n, temp, i = 0, j;

    int bin[32];

    printf("Enter a decimal number: ");

    scanf("%d", &n);

    printf("The octal value is: %o\n", n);

    printf("The hexadecimal value is: %X\n", n);

    temp = n;

    printf("The binary value is: ");

    if(temp == 0) {

        printf("0\n");

    } else {

        while(temp > 0) {

            bin[i] = temp % 2;

            temp = temp / 2;

            i++;

        }

        for(j = i - 1; j >= 0; j--) {

            printf("%d", bin[j]);

        }

        printf("\n");

    }

    return 0;

}

```

Output:

Enter a decimal number: 23

The octal value is: 27

The hexadecimal value is: 17

The binary value is: 10111

8. Write a C program to calculate difference between two time periods.

Program:

```
#include <stdio.h>
```

```
struct Time {
```

```
    int h;
```

```
    int m;
```

```
    int s;
```

```
};
```

```
int main() {
```

```
    struct Time t1, t2, diff;
```

```
    printf("Enter larger time period (hours minutes seconds): ");
```

```
    scanf("%d %d %d", &t1.h, &t1.m, &t1.s);
```

```
    printf("Enter smaller time period (hours minutes seconds): ");
```

```
    scanf("%d %d %d", &t2.h, &t2.m, &t2.s);
```

```

if(t2.s > t1.s) {
    t1.s = t1.s + 60;
    t1.m = t1.m - 1;
}
diff.s = t1.s - t2.s;
if(t2.m > t1.m) {
    t1.m = t1.m + 60;
    t1.h = t1.h - 1;
}
diff.m = t1.m - t2.m;
diff.h = t1.h - t2.h;

printf("The time difference is: %d hours %d minutes %d seconds\n", diff.h, diff.m,
diff.s);

return 0;
}

```

Output:

Enter larger time period (hours minutes seconds): 2

22

4

Enter smaller time period (hours minutes seconds): 1

34

56

The time difference is: 0 hours 47 minutes 8 seconds

9. Write a C program to implement an employee management system using structures.

Program:

Enter number of employees: 2

Enter ID, Name, and Salary for employee 1: 123

sam

23444

Enter ID, Name, and Salary for employee 2: 124

somu

24665

Employee Management System Details:

Employee ID: 123, Name: sam, Salary: 23444.00

Employee ID: 124, Name: somu, Salary: 24665.00

10. Write a C program to calculate area and perimeter of a rectangle and Triangle using function pointers.

Program:

```
#include <stdio.h>
```

```
float rectArea(float w, float h) {  
    return w * h;  
}
```

```
float rectPerim(float w, float h) {  
    return 2 * (w + h);  
}
```

```
}
```

```
float triArea(float b, float h) {  
    return 0.5 * b * h;  
}
```

```
float triPerim(float s1, float s2, float s3) {  
    return s1 + s2 + s3;  
}
```

```
int main() {  
    float (*rA)(float, float) = rectArea;  
    float (*rP)(float, float) = rectPerim;  
    float (*tA)(float, float) = triArea;  
    float (*tP)(float, float, float) = triPerim;  
    float rw, rh, tb, th, ts1, ts2, ts3;  
    printf("Enter width and height of rectangle: ");  
    scanf("%f %f", &rw, &rh);  
    printf("Enter base, height, and three sides of triangle: ");  
    scanf("%f %f %f %f %f", &tb, &th, &ts1, &ts2, &ts3);  
    printf("The Rectangle Area is: %.2f\n", rA(rw, rh));  
    printf("The Rectangle Perimeter is: %.2f\n", rP(rw, rh));  
    printf("The Triangle Area is: %.2f\n", tA(tb, th));  
    printf("The Triangle Perimeter is: %.2f\n", tP(ts1, ts2, ts3));  
    return 0;
```

}

Output:

Enter width and height of rectangle: 2

3

Enter base, height, and three sides of triangle: 3

6

2

3

5

The Rectangle Area is: 6.00

The Rectangle Perimeter is: 10.00

The Triangle Area is: 9.00

The Triangle Perimeter is: 10.00